



Urban Informatics

Assignment - Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 16

Total marks: (12 X 1) + (4 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

Which of the following statements are correct regarding JSON and CSV data formats?

- a. JSON stores data only in rows
- b. JSON supports nested and hierarchical structures
- c. CSV supports key–value pairs
- d. CSV preserves data types automatically

Correct Answer: b. JSON supports nested and hierarchical structures

Detailed explanation: Refer to lecture 21.

QUESTION 2 : [1 mark]

What problem does Object–Relational Mapping (ORM) primarily solve?

- a. Data visualization
- b. Manual SQL query writing
- c. Bridging database tables with programming objects
- d. Database installation

Correct Answer: c. Bridging database tables with programming objects

Detailed explanation: Refer to lecture 21.

QUESTION 3 : [1 mark]

Which Pandas attribute immediately reveals the dimensionality of a dataset?

- a. columns
- b. info()
- c. shape
- d. size

Correct Answer: c. shape

Detailed explanation: Refer to lecture 22.



QUESTION 4 : [1 marks]

Which feature makes SQLite suitable for lightweight applications?

- a. Requires dedicated server
- b. Embedded and serverless architecture
- c. Supports only NoSQL queries
- d. Operates only online

Correct Answer: b. Embedded and serverless architecture

Detailed explanation: Refer to lecture 23.

QUESTION 5 : [2 mark]

Match the following Visualization Techniques(Column I) with their most appropriate analytical use(Column II):

Column I	Column II
P. Histogram	i. Comparison of aggregated values across categories
Q. Boxplot	ii. Identification of distribution shape and frequency
R. Bar Plot	iii. Detection of outliers and comparison of spread
S. Kernel Density Plot	iv. Smooth estimation of probability density

- a. P–ii, Q–iii, R–i, S–iv
- b. P–iii, Q–ii, R–iv, S–i
- c. P–i, Q–iv, R–ii, S–iii
- d. P–iv, Q–i, R–iii, S–ii

Correct Answer: a. P–ii, Q–iii, R–i, S–iv

Detailed explanation: Refer to lecture 24.

QUESTION 6 : [1 mark]

Which plot is most effective for comparing distributions across categories?

- a. Line plot
- b. Scatter plot
- c. Boxplot
- d. Stem plot

Correct Answer: c. Boxplot

Detailed Solution: Refer to lecture 25.



QUESTION 7 : [1 mark]

Which rule is commonly used to flag outliers using quartiles?

- a. Z-score rule
- b. 3-sigma rule
- c. Tukey's rule
- d. Min–max rule

Correct Answer: c. Tukey's rule

Detailed Solution: Refer to lecture 25.

QUESTION 8 : [1 marks]

A DataFrame df contains a numeric column population with missing values and an index that is no longer meaningful after several row deletions. Which sequence of operations correctly fills missing values using estimation and then restores a clean default index?

- a. `df = df.reset_index()`
`df["population"] = df["population"].interpolate()`
- b. `df["population"] = df["population"].interpolate()`
`df = df.reset_index(drop=True)`
- c. `df = df.dropna()`
`df = df.set_index("population")`
- d. `df["population"] = df["population"].fillna(0)`
`df = df.reset_index()`

Correct Answer: b `df["population"] = df["population"].interpolate()`
`df = df.reset_index(drop=True)`

Detailed Solution: Refer to lecture 21.

Interpolate() estimates missing numeric values without removing rows, preserving data continuity. After row-level operations, **reset_index(drop=True)** restores a clean sequential index without retaining the old index as a column.



QUESTION 9 : [2 mark]

Match the following Data Inspection (Column I) & Preparation Operations (Column II)

Column I	Column II
P. info()	i. Provides statistical distribution of numerical variables
Q. describe()	ii. Displays schema, data types, and missing values
R. fillna()	iii. Replaces missing entries using a defined strategy
S. astype()	iv. Converts column data types explicitly

Choose the correct option:

- a. P-i, Q-ii, R-iv, S-iii
- b. P-ii, Q-i, R-iii, S-iv
- c. P-iv, Q-iii, R-i, S-ii
- d. P-iii, Q-iv, R-ii, S-i

Correct Answer: b. P-ii, Q-i, R-iii, S-iv

Detailed Solution: Refer to lecture 22.

QUESTION 10 : [1 mark]

A DataFrame df contains missing values in a numeric column area_sq_m.
Which approach is most suitable to fill missing values while minimizing outlier influence?

- a. df["area_sq_m"].fillna(df["area_sq_m"].mean())
- b. df["area_sq_m"].fillna(0)
- c. df["area_sq_m"].fillna(df["area_sq_m"].median())
- d. df.dropna()

Correct Answer: c. df["area_sq_m"].fillna(df["area_sq_m"].median())

Detailed Solution: Refer to lecture 22.

The median is a robust measure of central tendency and is less affected by extreme values or outliers. Filling missing values using the median preserves the overall distribution of the data better than the mean or constant replacement, making it suitable for skewed urban datasets.



QUESTION 11 : [1 mark]

Which of the following is the **primary functional advantage** of using an Object–Relational Mapping (ORM) framework compared to writing raw SQL queries?

- a. Faster execution in all scenarios
- b. Reduced memory usage
- c. Database abstraction and object-based interaction
- d. Automatic indexing of tables

Correct Answer: c. Database abstraction and object-based interaction

Detailed Solution: Refer to lecture 23.

QUESTION 12 : [2 mark]

Which code correctly visualizes the distribution and density of a continuous variable PM25?

- a. `df["PM25"].plot(kind="bar")`
- b. `sns.histplot(df["PM25"], kde=True)`
- c. `df.groupby("PM25").count().plot()`
- d. `plt.scatter(df.index, df["PM25"])`

Correct Answer: b. `sns.histplot(df["PM25"], kde=True)`

Detailed Solution: Refer to lecture 24.

A histogram visualizes the distribution of a continuous variable by grouping values into bins, while the kernel density estimate (KDE) provides a smooth approximation of the probability density. The **`sns.histplot()`** function with **`kde=True`** correctly displays both the frequency distribution and the underlying density of PM2.5 values.

QUESTION 13 : [1 mark]

Given a DataFrame `df` with columns `Area` and `PM25`, which code correctly computes and visualizes the mean PM2.5 concentration by urban area?

- a. `df.plot(x="Area", y="PM25", kind="bar")`
- b. `df.groupby("Area")["PM25"].mean().plot(kind="bar")`
- c. `df.groupby("PM25")["Area"].mean().plot()`
- d. `df["PM25"].mean().plot(kind="bar")`

Correct Answer: b. `df.groupby("Area")["PM25"].mean().plot(kind="bar")`



Detailed Solution: Refer to lecture 24.

To compute the mean PM2.5 concentration separately for each urban area, the data must first be grouped by Area. The `groupby("Area")["PM25"].mean()` operation performs the area-wise aggregation, and `plot(kind="bar")` correctly visualizes the aggregated values.

QUESTION 14 : [2 marks]

Match the following ORM Component /Object (Column I) with the Functional Responsibility (column II)

Column I	Column II
P. Declarative Model	i. Maps database tables to Python classes using schema definitions
Q. Session	ii. Manages transactional scope and persistence of ORM objects
R. SQLAlchemy Engine	iii. Establishes and manages low-level database connectivity
S. Query Builder	iv. Constructs object-oriented queries independent of database syntax

Choose the correct option:

- a. P–ii, Q–i, R–iii, S–iv
- b. P–i, Q–ii, R–iii, S–iv
- c. P–iii, Q–iv, R–i, S–ii
- d. P–iv, Q–iii, R–ii, S–i

Correct Answer: b. P–i, Q–ii, R–iii, S–iv

Detailed Solution: Refer to lecture 23.

QUESTION 15 : [1 mark]

Which code line correctly computes the Interquartile Range (IQR) of a numeric column values?

- a. `iqr = values.max() - values.min()`
- b. `iqr = values.quantile(0.75) - values.quantile(0.25)`
- c. `iqr = values.mean() - values.median()`
- d. `iqr = values.std()`

Correct Answers: b. `iqr = values.quantile(0.75) - values.quantile(0.25)`

Detailed Solution: Refer to lecture 25.

The interquartile range (IQR) measures the spread of the middle 50% of the data and is defined as the difference between the third quartile (Q3) and the first quartile (Q1). The **quantile(0.75)** and **quantile(0.25)** functions correctly compute Q3 and Q1, respectively.



QUESTION 16 : [1 mark]

A dataset contains a numeric column value with varying magnitudes. The objective is to rescale the column to a fixed range between 0 and 1, so that comparisons across variables become meaningful. Which code correctly performs this transformation?

- a. `df["value"] = df["value"] / df["value"].mean()`
- b. `df["value"] = (df["value"] - df["value"].min()) / df["value"].max()`
- c. `df["value"] = (df["value"] - df["value"].min()) / (df["value"].max() - df["value"].min())`
- d. `df["value"] = df["value"] / df["value"].std()`

Correct Answer: c `df["value"] = (df["value"] - df["value"].min()) / (df["value"].max() - df["value"].min())`

Detailed Solution: Refer to lecture 25.

Min–max normalization rescales values by subtracting the minimum and dividing by the data range, ensuring all values lie between 0 and 1.

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